Part A Chapter 11

Adaptive Behaviour[1]

1 Introduction

The fourth class of behavioural patterns addressed is adaptive behaviour. In this case, based on certain experiences, an individual's behaviour can change forever, or at least for longer lasting periods of time. Other terms sometimes used for specific types of adaptive behaviour are conditioning, or learned behaviour. The example discussed is learning to react on a wasp. If humans encounter a wasp, basically two types of attitudes can be shown. Some act in a relaxed manner, maybe trying not to be too close to the wasp, but without any panic reaction. Others may react in a completely different manner. They show behaviours like getting very nervous or even panicking, and trying to hit the wasp or immediately running away. This difference in behaviour, and, particularly, how the relaxed form of behaviour can transform to the panicking form, is one of the two types of adaptive behaviour that will be discussed in this chapter.

The other type of adaptive behaviour in this chapter explored in Section 6 is trust-based behaviour. Trust is the attitude an individual has with respect to the willingness to rely on some other individual, or object, or a turn of events. The individual might for example trust that the statements made by another individual are true. The individual might trust the commitment of another individual with respect to a certain (joint) goal. The individual might trust that another individual is capable of performing certain tasks, or an object that can be used for such a task (e.g., a car, a computer). Trust is considered as an internal cognitive state property of an individual that may help him to conduct various kinds of complex behaviour. By having such a cognitive

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function the individual may be able to cope with complex environments in which otherwise it would be impossible to behave in a proper way, by making decisions that have a good chance to be beneficial for the individual.

1.1 Domain description and focus

A first question is how it can be observed that an individual has adapted, i.e., has changed its behaviour. One way of addressing this question is to take (a set of) behaviour traces before the change and (a set of) behaviour traces after the change and, by comparing these, find a dynamic relationship in which these sets differ. For the first example this would look like this.

Before adaptation
If the individual observes that a wasp is close to the individual, then the individual will not panic.

After adaptation
If the individual observes that a wasp is close to the individual, then the individual will panic.

Notice that both relationships define stimulus-response behaviour, but not the same; therefore they indicate that adaptation has taken place from one type of stimulus-response behaviour to another type.

1.2 Characteristic patterns to be explored

Finding out such a distinction in dynamic relationships is useful. It defines in which respect the behaviour has changed over time. However, it does not give any information why, under which circumstances, such a change in behaviour has occurred or will occur. The above properties define the *result* of adaptation, but give no information about the *process* of adaptation itself. The next challenge is to find external dynamic relationships that define that a process of adaptation takes place. As an example, consider the traces shown in Table 1

time / *trace*	*time point 0*	*time point 1*	*time point 2*	*time point 3*	*time point 4*	*time point 5*	*time point 6*	*time point 7*	*time point 8*
trace 1	wasp close	wasp close	wasp close wasp stings	wasp close ouch!	wasp close panic	no wasp panic	no wasp	wasp close	wasp close panic
trace 2	wasp close	wasp close	no wasp	no wasp	wasp close	wasp close	no wasp	wasp close	wasp close
trace 3	wasp close	wasp close wasp stings	wasp close ouch!	no wasp panic	wasp close	wasp close panic	no wasp panic	wasp close	wasp close panic
trace 4	wasp close	wasp close	no wasp	wasp close	wasp close wasp stings	no wasp ouch!	no wasp	wasp close	wasp close panic

Table 1 Set of external behavioural traces for observable adaptive behaviour

From these example traces (and a number of similar ones) it seems that the criterion for adaptation (or *conditioning*) to take place is that the wasp stings. Or is a better criterion that the wasp stings and immediately after that the individual panics, as in trace 1 ? Among the example traces at least some traces are present in which the wasp stings and the individual does not panic. What will happen then, the next time the individual encounters a wasp ? In these traces (i.e., trace 4) the individual also panics if later on a wasp is close.

2 Analysis of the Main Aspects and Their Relations

In this section the main concepts and their relations needed for the model are identified.

2.1 Identification of relevant concepts

Among the concepts that play a central role in adaptive behaviour are observed *experiences* that have a longer term effect on behaviour. The most extreme forms of such experiences are *traumatic experiences*: they already have effect after one occurrence. Other types of experiences are milder and have their effect after a number of times; for example, this may hold for conditioning processes, such as the one described for *Aplysia* in Part A, Chapter 3.

Based on such experiences, via their *sensory representations* certain persisting internal cognitive *sensitivity* states are created, which connect to other internal cognitive states that, depending on such a sensitivity, generate behaviour in the form of *preparation for* and *performance of* certain *actions* as a response to certain observed *stimuli*. For example, for the wasp case the experiences are being stung by a wasp, a sensitivity state is the sensitivity to wasps, the stimuli involved are observing a wasp close by, and the action is to panic. Notice that for experiences it may be the case that they have their effect only in combination, for example, a tone (e.g., of a fridge that is opened) followed by food becoming available. Moreover, for a certain type of sensitivity, multiple sensitivity states can be used as intermediate states in the process of becoming sensitive: they represent gradual forms of sensitivity, that accumulates on the basis of a number of experiences. In summary, the following concepts are relevant:

- experiences
- sensitivity
- observation and sensory representations of stimuli and experiences
- preparations and performance of actions

2.2 Global structure of the model

The model is specified at two levels: the external behaviour level, and the internal, cognitive level. A global overall picture of the model can be found in Figure 1.

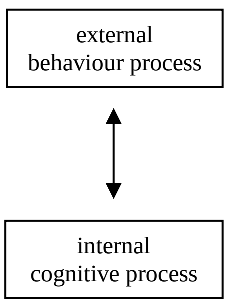

Figure 1 Global structure of the model

Interaction between the levels can take place by transferring input from the behavioural level to the cognitive level, determining output by the cognitive process, and transferring this output back to the behavioural level. Alternatively, the model at the behavioural level can also be used to determine the outputs from the inputs.

2.3 Relationships between the identified concepts

The relationships between the identified concepts are as follows; see also Figure 2.

Relationships at the external behavioural level for the wasp case

Viewed from an external behavioural perspective, the dynamic relationships between the main concepts are:

- experience of being stung affects panic
- experience of being stung affects screaming ouch!
- observing a wasp close affects panic

Relationships at the internal cognitive level for the wasp case

Viewed from an internal cognitive perspective, the dynamic relationships between the main concepts are:

- observation of being stung affects sensory representation of being stung
- sensory representation of being stung affects pain
- sensory representation of a wasp close affects sensitivity for wasps
- pain affects preparation for screaming ouch!
- pain affects sensitivity for wasps
- sensitivity for wasps affects preparation for panic
- observing a wasp close affects sensory representation of wasp being close
- sensory representation of wasp being close affects preparation for panic
- preparation for screaming ouch! affects screaming ouch!
- preparation for panic affects panic

Figure 2 depicts these relationships. Notice that here just four sensitivity states are included. They indicate different degrees of sensitivity, with S3 the highest and S0 (non-sensitive) the lowest. This number is arbitrary. For example, a one-step adaptation (traumatic experience) can be modelled by two sensitivity states S0 and S1.

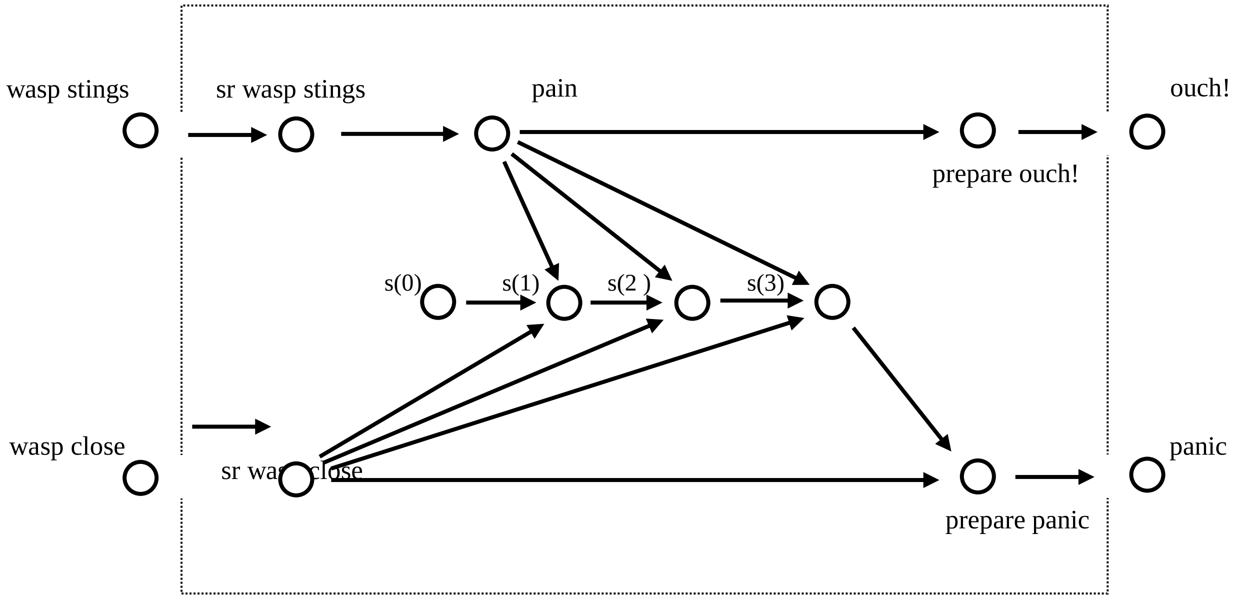

Figure 2 Relationships for the relevant concepts

3 The Detailed Model

In this section the detailed model is introduced. First it is shown in which way the relevant concepts are formalised, and next the dynamic relations are formalised in a detailed manner.

3.1 Formalising the concepts used

To obtain a detailed model, first the concepts used are formalised. In this case for each of the concepts a logical atom is introduced; see Table 2. These atoms are based on predicates for observation, sensory representations, preparations, action performance, experiences, sensitivity.

External behavioural level	
observation of stimuli (wasp close) of experiences (being stung)	 observed(wasp_close) observed(stung)
action (panic) (scream ouch!)	performed(panic) performed(ouch!)
Internal cognitive level	
sensory representation (of wasp, of being stung)	sr(wasp_close) sr(stung)
sensitivity (for wasp)	s(0) , s(1), ….
preparation (panic)	prepfor(panic)

(scream ouch!)	prepfor(ouch!)

Table 2 Formalisation of the relevant concepts

3.2 Formalising the relationships

The dynamics of these concepts involve temporal relationships, which are analysed in more detail below. They describe the basic parts of the process; see also Figure 2.

3.2.1 Relationships at the external behavioural level

For the external behavioural level, the following dynamic relationship holds.

EAB1

for all time points
if the individual observes that a wasp stings,
then for all later time points
 if the individual observes that a wasp is close to the individual,
 then the individual will panic

Note the conditional statement after the 'then'. Due to this conditional, the behaviour entailed by the property depends on (the interaction with) the external world. In traces where never a wasp is close again, not any particular behaviour is entailed by this property. These traces without further encounters give no information about whether or not the individual has adapted.

These properties can be slightly reformulated by taking all conditionals in once (but note the time indications; they are essential):

EAB2

for all time points
if the individual observes that a wasp is close to the individual,
 and at an earlier time point
 the individual observed that a wasp stung
then the individual will panic

To be complete, also the ouch! response is covered by the following dynamic relationship:

EAB3

for all time points
if the individual observes that a wasp stings,
then the individual will scream ouch!

Moreover, the behaviour if no stinging, and hence no adaptation has taken place has still to be specified:

EAB4

for all time points
if the individual observes that a wasp is close to the individual

and at no earlier point in time the individual observed that a wasp stung,
then the individual will not show panic

3.2.2 Relationships at the internal cognitive level

To realize adaptive behaviour, something (a cognitive state property) should change within the individual, and this change is to persist. The most simple way chosen here to obtain this is by assuming an internal cognitive state property s for sensitivity for wasps. However, to also take into account an internal pain state leading to the response ouch!, another internal cognitive state property pain is used (notice that this is not strictly needed for the ouch! response). The internal dynamics is illustrated by the traces from an internal cognitive perspective shown in Table 3, for the case of a one-step adaptation process. Notice that at some places at the time axis more time points are distinguished than for the external trace to make a difference between internal events. To keep the numbering of time points comparable to the external traces the numbering has been refined by using, for example, time points 3a and 3b. Moreoiver, to keep the table a bit simpler the sensory representations and preparations are not depicted.

time trace 1	*time point 0*	*time point 1*	*time point 2*	*time point 3a*	*time point 3b*	*time point 4*	*time point 5*	*time point 6*	*time point 7*	*time point 8*
input	wasp close	wasp close	wasp close wasp stings	wasp close	wasp close	wasp close	no wasp	no wasp	wasp close	wasp close
internal				pain	s(1)	s(1)	s(1)	s(1)	s(1)	s(1)
output					ouch!	panic	panic			panic
time trace 3	*time point 0*	*time point 1*	*time point 2a*	*time point 2b*	*time point 3*	*time point 4*	*time point 5*	*time point 6*	*time point 7*	*time point 8*
input	wasp close	wasp close wasp stings	wasp close	wasp close	no wasp	wasp close	wasp close	no wasp	wasp close	wasp close
internal			pain	s(1)	s(1)	s(1)	s(1)	s(1)	s(1)	s(1)
output				ouch!	panic		panic	panic		panic
time trace 4	*time point 0*	*time point 1*	*time point 2*	*time point 3*	*time point 4*	*time point 5a*	*time point 5b*	*time point 6*	*time point 7*	*time point 8*
input	wasp close	wasp close	no wasp	wasp close	wasp close wasp stings	wasp close	no wasp	no wasp	wasp close	wasp close
internal						pain	s(1)	s(1)	s(1)	s(1)
output							ouch!			panic

Table 3 Internal cognitive traces showing adaptive behaviour

The following internal dynamic relationships hold for these internal cognitive traces. Notice that for the sake of simplicity a one-step adaptation is described.

IAB1

at any point in time
if the individual observes someting, (e.g., wasp close, being stung)
then after this the corresponding sensory representation will be there

Formally

observed(X) $\twoheadrightarrow$ sr(X)

IAB2

at any point in time
if the sensory representation for being stung holds,
then after this pain holds

Formally

sr(stung) $\twoheadrightarrow$ pain

IAB3

at any point in time
if pain holds,
and the sensory representation of a wasp close holds
and s(0) holds
then after this s(1) holds

Formally

pain $\wedge$ sr(wasp_close) $\twoheadrightarrow$ s(1)

IAB4

at any point in time
if s(1) holds,
then at all later time points s(1) holds

Formally

s(1) $\twoheadrightarrow$ s(1)

IAB5

at any point in time
if s(1) holds,
and the individual has a sensory representation that a wasp is close
then the individual will prepare for panic

Formally

s(1) $\wedge$ sr(wasp_close) $\twoheadrightarrow$ prepfor(panic)

IAB6

at any point in time
if the individual is prepared for an action (e.g., panic, ouch!)
then the individual will perform this action

Formally

prepfor(X) $\twoheadrightarrow$ performed(X)

IAB7

at any point in time
if pain holds,
then the individual will prepare for screaming ouch!

Formally

pain $\twoheadrightarrow$ prepfor(ouch!)

Notice that the internal cognitive state property pain is not claimed to be persistent (the pain is just for a short time), whereas the internal cognitive state property s(1) has to be persistent in virtue of property **IAB4**. In graphical format, these properties are depicted in Figure 2 above.

4 Simulation Experiments

Using the LEADSTO software environment a number of simulation experiments have been performed, generating internal cognitive traces of the adaptation process. In Figure 3 an example of such an internal cognitive trace is shown (the other traces show a similar pattern). In this trace a scenario was taken, where five times a wasp comes close, but it only stings the third time. Besides the ouch! statement this leads to the sensitivity s(1), and also to (externally observable) panic behaviour. As the sensitivity s(1) persists over time, similarly the next times that a wasp comes close the panic occurs.

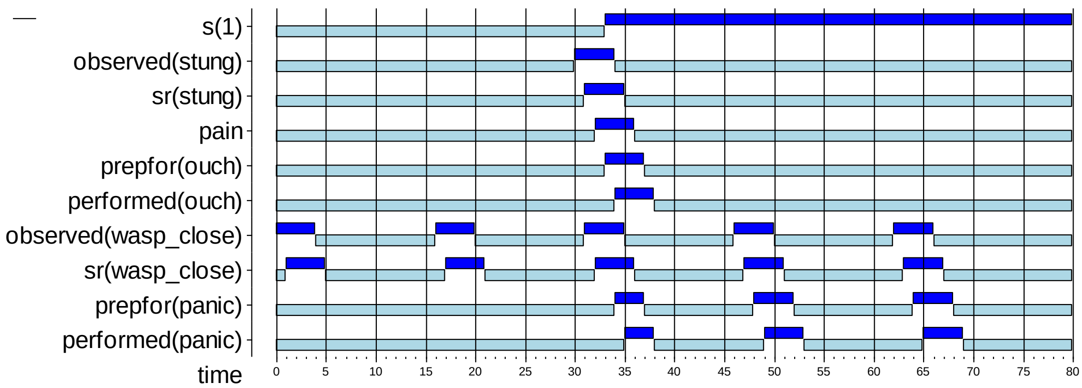

Figure 3 Simulation trace of a wasp sensitivity adaptation process

5 Evaluation

In this section, the simulations performed using the model are evaluated with respect to the characteristic patterns identified in Section 1.2. Moreover, the assumptions behind the model are summarised. Finally some possibilities for variations or refinements of the model are discussed.

5.1 Evaluation of the simulations and characteristic patterns

A number of different simulations conducted showed the characteristic patterns identified in Section 1.2. It is easy to perform a few more experiments to check more scenarios.

5.2 Assumptions made

The following assumptions were made.

- Precise timings of the steps were neglected.
- Sensitivity for wasps persists forever.
- Sensitivity is created by one experience only.
- Sensitivity for wasps has an all or nothing character (no gradual sensitivities).

5.3 Possible variations and refinements

Variants and refinements of the model can be made by taking into account the following possibilities.

- Incorporating precise timings of the steps.
- Sensitivity for wasps does not persist forever; extinction takes place.
- Sensitivity for wasps has a gradual character; it is acquired by a number of experiences, similar to how *Aplysia* acquires its sensitivity to siphon touches (see Part A, Chapter 4). In Figure 2 this possibility was already depicted.

6 Further Explorations: Trust-Based Behaviour

As an example of adaptive behaviour to be explored further, a simple example of trust-based behaviour is considered. Trust is a specific type of sensitivity that can be analysed referring to observations of experiences which in turn lead to expectations. For example, Lewis and Weigert (1985) state: '… observations that indicate that members of a system act according to and are secure in the expected futures constituted by the presence of each other for their symbolic representations'. Elofson (1998) defines trust as: 'trust is the outcome of observations leading to the belief that the actions of another may be relied upon, without explicit guarantee, to achieve a goal in a risky situation.' Elofson notes that trust can be developed over time as the outcome of a series of confirming observations. What is put forward is that one of the important factors from which trust results is the individual's history of experiences. As trust plays an important role in making decisions, the behaviour is adapted on the basis of the history of experiences.

Each event that can affect the degree of trust is interpreted by the individual to be either a *trust-negative experience* or a *trust-positive experience*. If the event is interpreted to be a trust-negative experience the individual will lose his trust to some degree, if it is interpreted to be trust-positive, the individual will gain trust to some degree. The degree to which the trust is changed may depend on the characteristics of the individual. This implies that the trusting individual performs a form of continual verification and validation of the subject of trust over time. For example, you can trust a car, based on a multitude of experiences with that specific car, and with other cars in general.

The example is in the context of buying fruit depending on the trust in the fruit quality in a specific shop close to your home. Sometimes, due to lack of time, you have to buy in this shop anyway. So, you will always continue to have experiences in this shop. At other times you have the time go to a shop further away. In these cases the decisions whether or not to avoid the nearer shop are made, based on your trust in the fruit of the shop nearby, which in turn is based your experiences (in comparison to the other shop). Examples of patterns that can be explored are:

- The buying behaviour depends on whether the experiences are mainly positive or mainly negative.
- Trust is increasing when experiences are positive, decreasing when they are negative.
- When there is less trust, more often the shop nearby will be avoided.

6.1 External Behavioural Dynamics

As possible *input* two types of *experience states* are assumed:

positive experience or *negative experience*

The possible *outputs* are:

to buy or *not to buy*

Note that here ‘to buy’ means that the person always go to that shop to buy. However, as the shop is by far the closest shop, even when the decision is ‘not to buy’, sometimes, when in a hurry the person will still go to the shop and may have more experiences.

6.2 Internal Cognitive Trust Dynamics

Four possible *internal cognitive trust states* are assumed:

- unconditional trust (ut)
- conditional trust (ct)
- conditional distrust (cd)
- unconditional distrust (ud)

The temporal relationships for the cognitive trust states are specified in executable format as follows:

if as input a positive experience is observed,
and the trust state is ud,
then the trust state will be cd, and not anymore ud

if as input a positive experience is observed,
and the trust state is cd,
then the trust state will be ct, and not anymore cd

if as input a positive experience is observed,
and the trust state is ct,

then the trust state will be ut, and not anymore ct

if as input a negative experience is observed,
and the trust state is cd,
then the trust state will be ud, and not anymore cd

if as input a negative experience is observed,
and the trust state is ct,
then the trust state will be cd, and not anymore ct

if as input a negative experience is observed,
and the trust state is ut,
then the trust state will be ct, and not anymore ut

if the trust state is ct,
then the individual will buy in the shop

if the trust state is ut,
then the individual will buy in the shop

if the trust state is cd,
then the individual will not buy in the shop

if the trust state is ud,
then the individual will not buy in the shop

All trust states are persistent, until they are changed by one of the specified rules. Notice that these functional role descriptions show many dependencies among the trust states, and that the decision to buy or not to buy may depend on very long histories; for this reason it is not easy to specify the behaviour from an external behaviourial perspective. As an example, starting with the state ct of conditional trust, a sequence of experiences

+ + − + − − + − −

leads to the subsequent trust states and avoid/not avoid decisions as depicted in the table below.

	time point 0	time point 1	time point 2	time point 3	time point 4	time point 5	time point 6	time point 7	time point 8	time point 9
input	+	+	-	+	-	-	+	-	-	+
internal	ct	ut	ut	ct	ut	ct	cd	ct	cd	ud
output		to buy	to buy	to buy	to buy	to buy	to buy	not to buy	to buy	not to buy

Here the trust state and the experience state at one time point determines the trust state of the next time point (at the right hand side). For example, at time point 2 the trust state is ut and the experience state is -. The resulting trust state in time point 3 (after update) is ct.

Explanation of trust-based behaviour

An explanation of trust-based behaviour from a cognitive perspective runs as follows:

Why does the individual at time point 8 buy?

The individual buys at time point 8, because it has a conditional trust at time point 7.

Why did the individual have conditional trust at time point 7?

The individual has conditional trust at time point 7, because it had a positive experience at time point 6, and at that time it had a conditional distrust.

Why did the individual have conditional distrust at time point 6?

The individual has conditional distrust at time point 6, because it had a negative experence at time point 5, and at that time it had a conditional trust.

……..

And so on, 8 steps back in time.

The following questions are relevant in this context:

- Can you give a trace of 20 experience states, so that if the first experience is changed into the opposite one, and the other experience states remain the same, the decision at time 20 also changes?
- Can the same be done for a trace of 1000 long?
- Do you think it is realistic and/or reasonable to count your experiences of a long time ago in your decisions?

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